

Лабораторная работа №1

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1. Цель работы

Цель данной лабораторной работы - приобретение практических навыков работы с системой управления версиями Git, языком программирования Julia.

2. Задание

1. Выполнить задания из лабораторной работы
2. Оформить отчет по лабораторной работе с помощью Markdown
3. Выполнить домашнюю работу

3. Теоретическое введение

Системы контроля версий (Version Control System, VCS) применяются при работе нескольких человек над одним проектом. Обычно основное дерево проекта хранится в локальном или удалённом репозитории, к которому настроен доступ для участников проекта. При внесении изменений в содержание проекта система контроля версий позволяет их фиксировать, совмещать изменения, произведённые разными участниками проекта, производить откат к любой более ранней версии проекта, если это требуется.

Системы контроля версий также могут обеспечивать дополнительные, более гибкие функциональные возможности. Например, они могут поддерживать работу с несколькими версиями одного файла, сохраняя общую историю изменений до точки ветвления версий и собственные истории изменений каждой ветви. Кроме того, обычно доступна информация о том, кто из участников, когда и какие изменения вносил. Обычно такого рода информация хранится в журнале изменений, доступ к которому можно ограничить.

1.2.1.4 Создание ключа ssh 1. Общая информация 1. Алгоритмы шифрования ssh 1. Аутентификация В SSH поддерживаются четыре алгоритма аутентификации по открытым ключам: — DSA: — размер ключей DSA не может превышать 1024, его следует отключить; — RSA: — следует создавать ключ большого размера: 4096 бит; — ECDSA: — ECDSA завязан на технологиях NIST, его следует отключить; — Ed25519: — используется пока не везде.

По умолчанию пользовательские ssh-ключи сохраняются в каталоге `~/.ssh` в домашнем каталоге пользователя.

Создание ключа ssh — Ключ ssh создаётся командой: `ssh-keygen -t`

1.2.2.5.1 Основные ветки (master) и ветки разработки (develop) Для фиксации истории проекта в рамках этого процесса вместо одной ветки master используются две ветки. В ветке master хранится официальная история релиза, а ветка develop предназначена для объединения всех функций. Кроме того, для удобства рекомендуется присваивать всем коммитам в ветке master номер версии.

1.2.2.5.2 Функциональные ветки (feature)

Под каждую новую функцию должна быть отведена собственная ветка, которую можно отправлять в центральный репозиторий для создания резервной копии или совместной работы команды. Ветки feature создаются не на основе master, а на основе develop. Когда работа над функцией завершается, соответствующая ветка сливается обратно с веткой develop. Функции не следует отправлять напрямую в ветку master. Как правило, ветки feature создаются на основе последней ветки develop.

2.1 Литературное (грамотное) программирование — это подход, приоритизирующий понятность программы для человека, а не её исполнение компьютером. В экосистеме Julia он реализуется через несколько инструментов. Предлагается использовать `Literate.jl`.

Идея, предложенная Дональдом Кнудом [1—3], меняет традиционный взгляд: программа создаётся не как последовательность инструкций для машины, а как литературное эссе, объясняющее логику решения задачи.

— `Literate.jl` отлично подходит для автоматизации создания документации, а также для написания статей и учебников прямо в `.jl`-файлах. — `Weave.jl` больше заточен под академические отчёты и публикации, требующие строгого форматирования (LaTeX/PDF). — `Jupyter` лучше подходит для интерактивного исследования данных, но управление версиями `.ipynb`-файлов и их автоматизация сложнее.

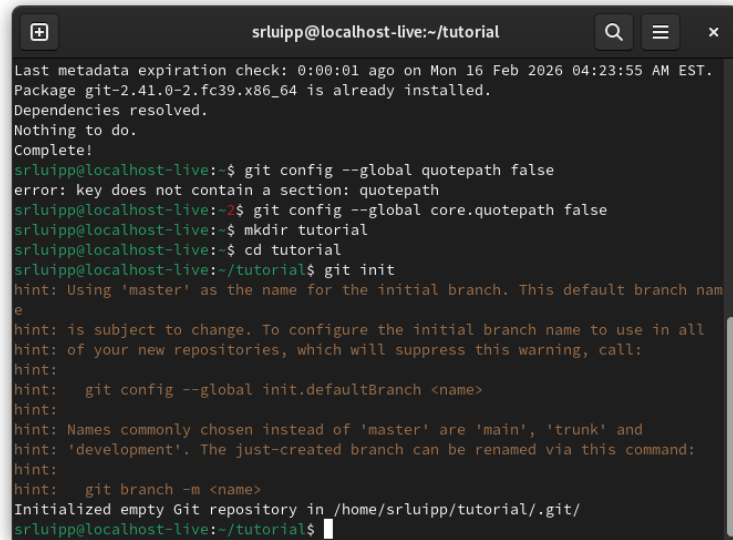
4. Выполнение лабораторной работы

Начинаю с подготовки и настройки рабочего пространства (установка git) (рис. 1).

```
logout
srluipp@localhost-live:~$ git config --global user.name "srluipp"
srluipp@localhost-live:~$ git config --global user.email "1132236039@gmail.com"
srluipp@localhost-live:~$ git config --global quotepath false
error: key does not contain a section: quotepath
srluipp@localhost-live:~$ sudo dnf install git
[sudo] password for srluipp:
Fedora 39 - x86_64 [ === ] --- B/s | 0 B --:-- ETA
```

Рисунок 1: git1

Настраиваю git configs(рис. 2).



```
srluipp@localhost-live:~/tutorial
Last metadata expiration check: 0:00:01 ago on Mon 16 Feb 2026 04:23:55 AM EST.
Package git-2.41.0-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
srluipp@localhost-live:~$ git config --global quotepath false
error: key does not contain a section: quotepath
srluipp@localhost-live:~$ git config --global core.quotepath false
srluipp@localhost-live:~$ mkdir tutorial
srluipp@localhost-live:~$ cd tutorial
srluipp@localhost-live:~/tutorial$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/srluipp/tutorial/.git/
srluipp@localhost-live:~/tutorial$
```

Рисунок 2: gitconfigs

Пробую делать коммиты(рис. 3).

```
Initialized empty Git repository in /home/srluipp/tutorial/.git/
srluipp@localhost-live:~/tutorial$ echo 'hello world' > hello.txt
srluipp@localhost-live:~/tutorial$ git add hello.txt
srluipp@localhost-live:~/tutorial$ git commit -am 'Новый файл'
[master (root-commit) c8b5857] Новый файл
1 file changed, 1 insertion(+)
create mode 100644 hello.txt
```

Рисунок 3: commits

Настраиваю gitignores(рис. 4) (рис. 5).

```

srluipp@localhost-live:~/tutorial$ curl -L -s https://www.gitignore.io/api/list
1c,1c-bitrix,a-frame,actionscript,ada
adobe,advancedinstaller,adventuregamestudio,agda,al
alteraquartusii,altium,amplify,android,androidstudio
angular,anjuta,ansible,ansibletower,apachecordova
apachehadoop,appbuilder,appcelerator titanium,appcode,appcode+all
appcode+iml,appengine,aptanastudio,arcanist,archive
archives,archlinuxpackages,asdf,aspnetcore,assembler
astro,ate,atmelstudio,ats,audio
autohotkey,automationstudio,autotools,autotools+strict,awr
azurefunctions,azurite,backup,ballerina,basercms
basic,batch,bazaar,bazel,bitrise
bitrix,bittorrent,blackbox,blender,bloop
bluej,bookdown,bower,bricxcc,buck
c,c++,cake,cakephp,cakephp2
cakephp3,calabash,carthage,certificates,ceylon
cfwheels,cheffcookbook,chocolatey,circuitpython,clean
clion,clion+all,clion+iml,clojure,cloud9
cmake,cocoapods,cocos2dx,cocoscreator,codeblocks
codecomposerstudio,codeigniter,codeio,codekit,codesniffer
coffeescript,commonlisp,compodoc,composer,compressed
compressedarchive,compression,conan,concrete5,coq

```

Рисунок 4: gitignores

```

srluipp@localhost-live:~/tutorial$ curl -L -s https://www.gitignore.io/api/c >>
.gitignore
srluipp@localhost-live:~/tutorial$ curl -L -s https://www.gitignore.io/api/c++ >>
> .gitignore

```

Рисунок 5: gitignores2

Далее работаю с настройкой ключей ssh-keygen (рис. 6).

```

srluipp@localhost-live:~/tutorial$ ssh-keygen -C "srluipp <1132236039@pfur.ru>"
Generating public/private rsa key pair.
Enter file in which to save the key (/home/srluipp/.ssh/id_rsa):
/home/srluipp/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/srluipp/.ssh/id_rsa
Your public key has been saved in /home/srluipp/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:oT0smRw0YjPipjUqv50uDAQgvmv6J0kiZbw/40Kgr64 srluipp <1132236039@pfur.ru>
The key's randomart image is:
+---[RSA 3072]-----+
|+                |
|+                |
|o.= . . .       |
|.+. + B .       |
|. =o   B S       |
|+++ . .         |
|B0.o            |
|@+...           |
|EO==.           |
+---[SHA256]-----+

```

Рисунок 6: ssh

Получаю ключ rsa (рис. 7).

```

srluipp@localhost-live:~/tutorial$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDz4sH/aYQPZnio0r2li+Dsg7QFRwe+rWAGEpJVSsNb
TYiqtPAK/ub35D8zkvAQ/sAN9yX1B/qI8Ewi704EfWUFRap7RyW3At7ovovxoapi4nKLzqDuoJ6NQ6h+
4kIbVqwdTD9hUqqrAFyFufZeFvdPwly7WE0bc2tMxS5yG8pvcCs86hVGlw8nlv/RpvHtzZh0nI59vb57
4+c4b8JLiGz2oJzmHKm7XR02xS0KviP0SlnQHUPyKPU2g8pvSIyoGB2EWDguzcJ8hadEapLJAurmrmg+
yHwW/sRUJgHHj0e3YYrgCZcuX/KfN0XdMazdIi2OUfKWazmatjyNK7CDFesraoToD8067a6p6RNLJgqU
GB7D06+ykuEzP6/P7no0f0m05KdHXwUi0lF6EowftKB+P/tsYHVCKunazZEK+HtEWb9zQCl9IDpXbqHA
M9kM4GfIkv6AiyEd5VbJSdk5DW8z9srnPSdFaid5agoD4KILR60V/ZZe0qG9N12NRfgGoWE= srluipp
<1132236039@pfur.ru>

```

Рисунок 7: rsa

Работаю с ветками git master, git origin (рис. 8).

```

srluipp@localhost-live:~/tutorial$ git remote add origin git@github.com:srluipp/
study_2025-2026_matmod.git
srluipp@localhost-live:~/tutorial$ git push -u origin master
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 241 bytes | 241.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/srluipp/study_2025-2026_matmod/pull/new/master
remote:
To github.com:srluipp/study_2025-2026_matmod.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

Рисунок 8: git

Генерирую ключ gpg (рис. 9).

```

srluipp@localhost-live:~/tutorial2$ gpg --full-generate-key
gpg (GnuPG) 2.4.3; Copyright (C) 2023 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
  (1) RSA and RSA
  (2) DSA and Elgamal
  (3) DSA (sign only)
  (4) RSA (sign only)
  (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
    0 = key does not expire
    <n>  = key expires in n days
    <n>w  = key expires in n weeks
    <n>m  = key expires in n months
    <n>y  = key expires in n years

```

Рисунок 9: gpg

Устанавливаю необходимые параметры для ключа: 4096, rsa, key does not expire at all (0) (рис. 10) (рис. 11)..

```

(14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
    0 = key does not expire
    <n> = key expires in n days
    <n>w = key expires in n weeks
    <n>m = key expires in n months
    <n>y = key expires in n years
Key is valid for? (0) 0
Key does not expire at all
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: srluipp
Email address: 1132236039@pfur.ru
Comment: no
You selected this USER-ID:
    "srluipp (no) <1132236039@pfur.ru>"

```

Рисунок 10: gpg1

```

You selected this USER-ID:
    "srluipp (no) <1132236039@pfur.ru>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/srluipp/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/srluipp/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/srluipp/.gnupg/openpgp-revocs.d/F74
4298471BFE80C5F9B21E5AF422ED4C47CC4C7.rev'
public and secret key created and signed.

pub  rsa4096 2026-02-16 [SC]
     F744298471BFE80C5F9B21E5AF422ED4C47CC4C7
uid          srluipp (no) <1132236039@pfur.ru>
sub  rsa4096 2026-02-16 [E]

```

Рисунок 11: gpg2

Затем копирую ключ и прикрепляю новый gpg ключ на свой профиль в GitHub (рис. 12) (рис. 13)..

```

srluipp@localhost-live:~/tutorial$ gpg --list-secret-keys --keyid-format LONG
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
[keyboard]
-----
sec  rsa4096/AF422ED4C47CC4C7 2026-02-16 [SC]
     F744298471BFE80C5F9B21E5AF422ED4C47CC4C7
uid          [ultimate] srluipp (no) <1132236039@pfur.ru>
ssb  rsa4096/8715382E82C490FF 2026-02-16 [E]

```

Рисунок 12: gpg3

```

erluipp@localhost-live:~/tutorial$ gpg --armor --export F744298471BFE80C5F9B21E5
AF422ED4C47CC4C7
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQINBgmS5cUBEADCrnuddQn52VdBDNUw4AXSoC2vp5Itsv2zzxwGiupok1B0Ah5U
QpgD0D7qEga3rCdxgG63K9Xgn95MRZB/aBNYxweh+aueNz6u2D4JFVImAtSvxw+R
qgUPJAYBQ01mrRzx1TkW0U0/EPNefAnjeZH2V4eVrWh2IoJ2Kso51KMZXmSi5zSg
TtcLWi02/l0NzvkiiywEgK+53RawCfJAFc8agrVYif0vTSopeeRTrfNN7SREAcJ
kcXLh270FUzaT2xCRhZcuDU14bMkY/WBvBWIQoMUCyYjYfdC4UH5A6m2vhToVr2ki
m89u0sZkFi/Zcsh8GITIWIcV+Ib46MQyuJkHYy8TKmLE9c/+VCmv6D2HR09I7IYm
LDH+i4XjL80txc6SLnRTJp0GQyjm0k9bieoWze2SRY7N6x+YPhG9cUQn3ccutdJs
CXiBdkaraHDMVEbWwj095l9PMenuzCvuPLWyUbn0iWuf5JmKAPbzBYMBvY8GLBGC
+uLTFIWI7xR+3Q2lIsVmnMr3SPPKrNck3IY8HaPC5VNxtZyqJdNjMgtIwVt7X/0
yi+Px/VimGj3xp3WirN68BZoM0yE6gQ307eZjlsZ14TYxkxk8ySLL352HIVhnC2y
mauJcU85kNuHYowxtz2sfnKpaQDk/v8BsFGdRnTx0r2NZcHAo8ajC3qQ2QARAQAB
tCFZcmx1aXBwIChubykgPDEzMzIyMzYwMzIACGZ1c15ydT6JALEEEwEiADsWIQT3
RCmEcB/oDF+bIeWVQi7UxHzExwUCaZLlxQIBAwULCQgHAgIiAgYVCgkICwIEFgID
AQIEBwIXgAAKCRcVQi7UxHzEx032D/odH7ZtJ0mWacLBc0NXaKY3R/v703yZ/Nvr
GB8vLkW497xtn4YBZ8fHoF0RTU8K+klr4pusP3G+pnLwD/lyDR7qRxtD+76sYbd/
R+3f6uIeJDhWHZA5MFCtGrJkVEIzi7XjT1CKgmrBfir0jv1EK5qUCr60i8xBvyx1
0+Vkpqr2CjDnciT+SBM9A+d6rTlurYcqMbjL6LysFwL5mIfy60QGnid/LPWTPr2
xdsUKJeGQuQtQcDvdgPpsnE3pQBxtXNnXslcGT5PU4oWTFmc6GMe6UVAMdXknLH
F4GnnSUz1ow+Hu8PaQH1F34RCMFUmZkXHmUzahZbRcGAXRvb7S7XmGJvc6UwEMR

```

Рисунок 13: gpg4

Настраиваю configs (рис. 14).

```

erluipp@localhost-live:~/tutorial$ git config --global user.signingkey F74429847
1BFE80C5F9B21E5AF422ED4C47CC4C7
erluipp@localhost-live:~/tutorial$ git config --global commit.gpgsign true
erluipp@localhost-live:~/tutorial$ git config --global gpg.program $(which gpg)

```

Рисунок 14: configs2

Теперь необходимо установить и распаковать gitflow (рис. 15) (рис. 16) (рис. 17).

```

Complete!
erluipp@localhost-live:~/tutorial$ sudo python3 -m pip install gitflow
Collecting gitflow
  Downloading gitflow-0.5.1.zip (180 kB)
    180.2/180.2 kB 476.0 kB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done

```

Рисунок 15: gitflow1

```

srluipp@localhost-live:~$ wget -O git-flow.tar.gz https://github.com/petervanderdoes/gitflow/archive/develop.tar.gz
--2026-02-16 05:06:49-- https://github.com/petervanderdoes/gitflow/archive/develop.tar.gz
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/petervanderdoes/gitflow-avh/archive/develop.tar.gz [following]
--2026-02-16 05:06:50-- https://github.com/petervanderdoes/gitflow-avh/archive/develop.tar.gz
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/petervanderdoes/gitflow-avh/tar.gz/refs/heads/develop [following]
--2026-02-16 05:06:50-- https://codeload.github.com/petervanderdoes/gitflow-avh/tar.gz/refs/heads/develop
Resolving codeload.github.com (codeload.github.com)... 140.82.121.9
Connecting to codeload.github.com (codeload.github.com)|140.82.121.9|:443... connected.

```

Рисунок 16: gitflow2

```

install -d -m 0755 /usr/local/share/doc/gitflow/hooks
install -m 0755 git-flow /usr/local/bin
install -m 0644 git-flow-init git-flow-feature git-flow-bugfix git-flow-hotfix git-flow-release git-flow-support git-flow-version git-flow-log git-flow-config gitflow-common gitflow-shFlags /usr/local/bin
install -m 0644 hooks/filter-flow-hotfix-finish-tag-message hooks/filter-flow-hotfix-start-version hooks/filter-flow-release-branch-tag-message hooks/filter-flow-release-finish-tag-message hooks/filter-flow-release-start-version hooks/post-flow-bugfix-delete hooks/post-flow-bugfix-finish hooks/post-flow-bugfix-publish hooks/post-flow-bugfix-pull hooks/post-flow-bugfix-start hooks/post-flow-bugfix-track hooks/post-flow-feature-delete hooks/post-flow-feature-finish hooks/post-flow-feature-publish hooks/post-flow-feature-pull hooks/post-flow-feature-start hooks/post-flow-feature-track hooks/post-flow-hotfix-delete hooks/post-flow-hotfix-finish hooks/post-flow-hotfix-publish hooks/post-flow-hotfix-start hooks/post-flow-release-branch hooks/post-flow-release-delete hooks/post-flow-release-finish hooks/post-flow-release-publish hooks/post-flow-release-start hooks/post-flow-release-track hooks/pre-flow-feature-delete hooks/pre-flow-feature-finish hooks/pre-flow-feature-publish hooks/pre-flow-feature-pull hooks/pre-flow-feature-start hooks/pre-flow-feature-track hooks/pre-flow-hotfix-delete hooks/pre-flow-hotfix-finish hooks/pre-flow-hotfix-publish hooks/pre-flow-hotfix-start hooks/pre-flow-release-branch hooks/pre-flow-release-delete hooks/pre-flow-release-finish hooks/pre-flow-release-publish hooks/pre-flow-release-start hooks/pre-flow-release-track /usr/local/share/doc/gitflow/hooks
srluipp@localhost-live:~/gitflow-avh-develop$ git flow version
1.12.4-dev0 (AVH Edition)
srluipp@localhost-live:~/gitflow-avh-develop$

```

Рисунок 17: gitflow3

Настройка branches (рис. 18).

```

hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/srluipp/.git/
No branches exist yet. Base branches must be created now.
Branch name for production releases: [master] master
Branch name for "next release" development: [develop] develop

How to name your supporting branch prefixes?
Feature branches? [feature/] feature
Bugfix branches? [bugfix/] bugfix
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/srluipp/.git/hooks]
srluipp@localhost-live:~$

```


Рисунок 18: branches

Checking the branches (рис. 19) (рис. 20).

```
sruiipp@localhost-live:~$ git branch
* develop
  master
sruiipp@localhost-live:~$
```

Рисунок 19: branches2

```
sruiipp@localhost-live:~$ git flow feature start feature_branch
Switched to a new branch 'featurefeature_branch'

Summary of actions:
- A new branch 'featurefeature_branch' was created, based on 'develop'
- You are now on branch 'featurefeature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

sruiipp@localhost-live:~$ git flow feature finish feature_branch
Switched to branch 'develop'
Already up to date.
Deleted branch featurefeature_branch (was 9216a5c).

Summary of actions:
- The feature branch 'featurefeature_branch' was merged into 'develop'
- Feature branch 'featurefeature_branch' has been locally deleted
- You are now on branch 'develop'

sruiipp@localhost-live:~$
```

Рисунок 20: branches3

Так же необходимо предустановить gh (рис. 21).

```
sruiipp@localhost-live:~$ sudo dnf -y install gh
Last metadata expiration check: 0:52:21 ago on Mon 16 Feb 2026 04:23:55 AM EST.
Dependencies resolved.
=====
Package           Architecture Version           Repository        Size
=====
Installing:
gh                x86_64          2.45.0-1.fc39    updates          9.1 M
Transaction Summary
=====
Install 1 Package

Total download size: 9.1 M
Installed size: 46 M
Downloading Packages:
[MIRROR] gh-2.45.0-1.fc39.x86_64.rpm: Curl error (28): Timeout was reached for http://fedora-archive.ip-connect.vn.ua/fedora/linux/updates/39/Everything/x86_64/Packages/g/gh-2.45.0-1.fc39.x86_64.rpm [Failed to connect to fedora-archive.ip-connect.vn.ua port 80 after 30000 ms: Timeout was reached]
gh-2.45.0-1.fc39.x86_64. 0% [ ] 5.2 kB/s | 23 kB 29:49 ETA
```

Рисунок 21: gh

Установленный gitflow (рис. 22).

```

install -d -m 0755 /usr/local/share/doc/gitflow/hooks
install -m 0755 git-flow /usr/local/bin
install -m 0644 git-flow-init git-flow-feature git-flow-bugfix git-flow-hotfix git-flow-
release git-flow-support git-flow-version git-flow-log git-flow-config gitflow-common gi
tflow-shFlags /usr/local/bin
install -m 0644 hooks/filter-flow-hotfix-finish-tag-message hooks/filter-flow-hotfix-sta
rt-version hooks/filter-flow-release-branch-tag-message hooks/filter-flow-release-finish
-tag-message hooks/filter-flow-release-start-version hooks/post-flow-bugfix-delete hooks
/post-flow-bugfix-finish hooks/post-flow-bugfix-publish hooks/post-flow-bugfix-pull hook
s/post-flow-bugfix-start hooks/post-flow-bugfix-track hooks/post-flow-feature-delete hoo
ks/post-flow-feature-finish hooks/post-flow-feature-publish hooks/post-flow-feature-pull
hooks/post-flow-feature-start hooks/post-flow-feature-track hooks/post-flow-hotfix-dele
te hooks/post-flow-hotfix-finish hooks/post-flow-hotfix-publish hooks/post-flow-hotfix-s
tart hooks/post-flow-release-branch hooks/post-flow-release-delete hooks/post-flow-relea
se-finish hooks/post-flow-release-publish hooks/post-flow-release-start hooks/post-flow-
release-track hooks/pre-flow-feature-delete hooks/pre-flow-feature-finish hooks/pre-flow
-feature-publish hooks/pre-flow-feature-pull hooks/pre-flow-feature-start hooks/pre-flow
-feature-track hooks/pre-flow-hotfix-delete hooks/pre-flow-hotfix-finish hooks/pre-flow-
hotfix-publish hooks/pre-flow-hotfix-start hooks/pre-flow-release-branch hooks/pre-flow-
release-delete hooks/pre-flow-release-finish hooks/pre-flow-release-publish hooks/pre-fl
ow-release-start hooks/pre-flow-release-track /usr/local/share/doc/gitflow/hooks
sruiipp@localhost-live:~/gitflow-avh-develop$ git flow version
1.12.4-dev0 (AVH Edition)
sruiipp@localhost-live:~/gitflow-avh-develop$

```

Рисунок 22: gitflow

По заданию нужно установить nodejs (рис. 23) (рис. 24) (рис. 25).

```

root@localhost-live:~
File Edit View Search Terminal Help
Installing:
  npm      noarch      8.12.0-1.fc39      updates      2.6 M
  yarn      noarch      1.22.22-1          yarn         1.2 M
Installing dependencies:
  nodejs    x86_64      1:20.17.0-1.fc39   updates      51 k
  nodejs-libs x86_64      1:20.17.0-1.fc39   updates      15 M
Installing weak dependencies:
  nodejs-docs noarch      1:20.17.0-1.fc39   updates      8.3 M
  nodejs-full-i18n x86_64      1:20.17.0-1.fc39   updates      8.4 M
  nodejs-npm    x86_64      1:10.8.2-1.20.17.0.1.fc39 updates      2.1 M

Transaction Summary
=====
Install 7 Packages

Total size: 38 M
Total download size: 3.8 M
Installed size: 203 M
Downloading Packages:
[SKIPPED] nodejs-20.17.0-1.fc39.x86_64.rpm: Already downloaded
[SKIPPED] nodejs-docs-20.17.0-1.fc39.noarch.rpm: Already downloaded
[SKIPPED] nodejs-full-i18n-20.17.0-1.fc39.x86_64.rpm: Already downloaded
[SKIPPED] nodejs-libs-20.17.0-1.fc39.x86_64.rpm: Already downloaded

```

Рисунок 23: nodejs

```
root@localhost-live:~  
File Edit View Search Terminal Help  
REDHAT_BUGZILLA_PRODUCT_VERSION=39  
REDHAT_SUPPORT_PRODUCT_VERSION=39  
[root@localhost-live ~]# sudo dnf --installroot=/mnt/big --releasever=39 --setopt=reposd  
ir=/etc/yum.repos.d --nogpgcheck install -y nodejs npm  
Fedora 39 - x86_64 3.4 MB/s | 89 MB 00:26  
Fedora 39 openh264 (From Cisco) - x86_64 834 B/s | 2.6 kB 00:03  
Fedora 39 - x86_64 - Updates 343 kB/s | 42 MB 02:05  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repo	Size
Installing:				
nodejs	x86_64	1:20.17.0-1.fc39	updates	51 k
nodejs-npm	x86_64	1:10.8.2-1.20.17.0.1.fc39	updates	2.1 M
Installing dependencies:				
alternatives	x86_64	1.26-1.fc39	updates	39 k
audit-libs	x86_64	3.1.5-1.fc39	updates	123 k
authselect	x86_64	1.4.3-1.fc39	fedora	149 k
authselect-libs	x86_64	1.4.3-1.fc39	fedora	249 k
basesystem	noarch	11-18.fc39	fedora	7.2 k
bash	x86_64	5.2.26-1.fc39	updates	1.8 M
bzip2-libs	x86_64	1.0.8-16.fc39	fedora	41 k
ca-certificates	noarch	2024.2.69.v8.0.401-1.0.fc39	updates	871 k

Рисунок 24: nodejs2

```
srliupp@localhost-live:~$ sudo dnf -y install nodejs  
Last metadata expiration check: 1:57:48 ago on Mon 16 Feb 2026 04:23:55 AM EST.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
Installing:				
nodejs	x86_64	1:20.17.0-1.fc39	updates	51 k
Installing dependencies:				
nodejs-libs	x86_64	1:20.17.0-1.fc39	updates	15 M
Installing weak dependencies:				
nodejs-docs	noarch	1:20.17.0-1.fc39	updates	8.3 M
nodejs-full-i18n	x86_64	1:20.17.0-1.fc39	updates	8.4 M
nodejs-npm	x86_64	1:10.8.2-1.20.17.0.1.fc39	updates	2.1 M

Рисунок 25: nodejs3

Устанавливаю libxcrypt (рис. 26).

```
srliupp@localhost-live:~  
File Edit View Search Terminal Help  
srliupp@localhost-live:~$ sudo dnf -y install libxcrypt-compat  
Last metadata expiration check: 1:55:30 ago on Mon 16 Feb 2026 04:23:55 AM EST.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
libxcrypt-compat	x86_64	4.4.36-2.fc39	fedora	90 k

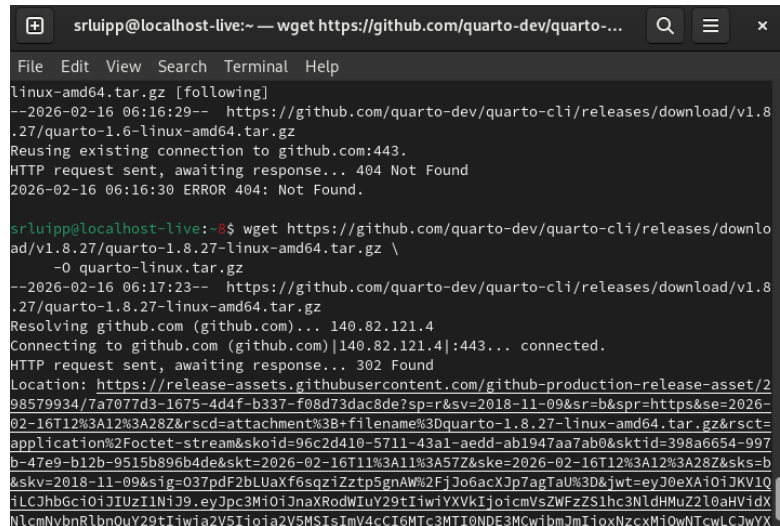
Transaction Summary
=====

Install 1 Package

Total download size: 90 k
Installed size: 193 k
Downloading Packages:
[MIRROR] libxcrypt-compat-4.4.36-2.fc39.x86_64.rpm: Curl error (28): Timeout was reached for http://fedora-archive.ip-connect.vn.ua/fedora/linux/releases/39/Everything/x86_64/os/Packages/l/libxcrypt-compat-4.4.36-2.fc39.x86_64.rpm [Failed to connect to fedora-archive.ip-connect.vn.ua port 80 after 30013 ms: Timeout was reached]
[MIRROR] libxcrypt-compat-4.4.36-2.fc39.x86_64.rpm: Curl error (28): Timeout was reached for https://dl.fedoraproject.org/pub/archive/fedora/linux/releases/39/Everything/x86_64/os/Packages/l/libxcrypt-compat-4.4.36-2.fc39.x86_64.rpm [Operation too slow. Less than

Рисунок 26: *libxscript*

Установка quarto (рис. 27).

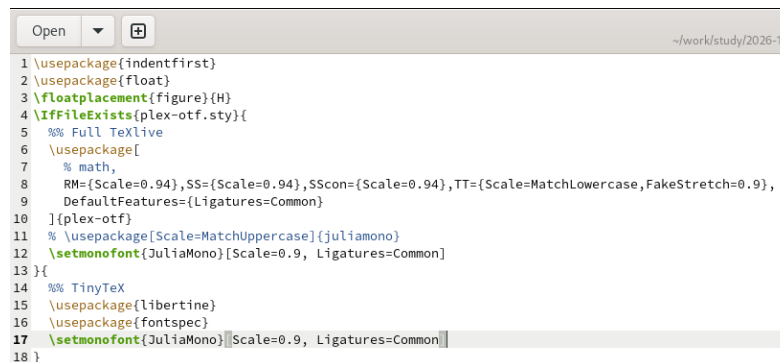


```
srluipp@localhost-live:~ — wget https://github.com/quarto-dev/quarto-...
File Edit View Search Terminal Help
linux-amd64.tar.gz [following]
--2026-02-16 06:16:29-- https://github.com/quarto-dev/quarto-cli/releases/download/v1.8
.27/quarto-1.6-linux-amd64.tar.gz
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 404 Not Found
2026-02-16 06:16:30 ERROR 404: Not Found.

srluipp@localhost-live:~$ wget https://github.com/quarto-dev/quarto-cli/releases/downlo
ad/v1.8.27/quarto-1.8.27-linux-amd64.tar.gz \
-O quarto-linux.tar.gz
--2026-02-16 06:17:23-- https://github.com/quarto-dev/quarto-cli/releases/download/v1.8
.27/quarto-1.8.27-linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)[140.82.121.4]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/2
98579934/7a7077d3-1675-4d4f-b337-f08d73dac8de?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-
02-16T12%3A12%3A28Z&rscd=attachment%3B+filename%3Dquarto-1.8.27-linux-amd64.tar.gz&rsct=
application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997
b-47e9-b12b-9515b896b4de&skt=2026-02-16T11%3A11%3A57Z&ske=2026-02-16T12%3A12%3A28Z&sk=b
&skv=2018-11-09&sig=037pdfF2bLUAxf6sqziZztp5gnAW%2FjJo6acXJp7agTaU%3D&jwt=eyJ0eXA10iJKV1Q
iLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiOiJjoicmVsZWZzS1hc3NldHMuzZ22L0aHVidX
NlcmVubnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MTI0NDU3MCwibmJmIjozNzcwMjQwNTcwLjYwYX
```

Рисунок 27: *quarto*

Добавила в преамбуле preamble.tex пакет juliamono (рис. 28) (рис. 29).



```
Open ~\work\study\2026-1/2
1 \usepackage{indentfirst}
2 \usepackage{float}
3 \floatplacement{figure}{H}
4 \ifFileExists{plex-otf.sty}{
5   %% Full TeXlive
6   \usepackage[
7     % math,
8     RM={Scale=0.94},SS={Scale=0.94},SScon={Scale=0.94},TT={Scale=MatchLowercase,FakeStretch=0.9},
9     DefaultFeatures={Ligatures=Common}
10  ]{plex-otf}
11 % \usepackage[Scale=MatchUppercase]{juliamono}
12 \setmonofont{JuliaMono}[Scale=0.9, Ligatures=Common]
13 }{
14   %% TinyTeX
15   \usepackage{libertine}
16   \usepackage{fontspec}
17   \setmonofont{JuliaMono}[Scale=0.9, Ligatures=Common]
18 }
```

Рисунок 28: *juliamono*

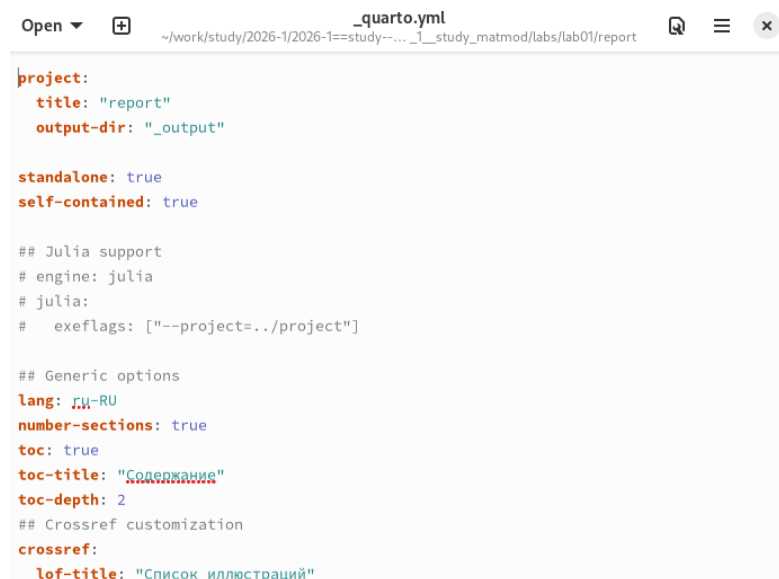
```
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2...
http://127.0.0.1:8888/lab?token=ad93ce8fb771a0c822d4cfbbf090f3483eba66c46e
89f643
Shut down this Jupyter server (y/[n])? y
[C 2026-02-17 16:21:03.276 ServerApp] Shutdown confirmed
[I 2026-02-17 16:21:03.277 ServerApp] Shutting down 5 extensions
[I 2026-02-17 16:21:03.278 ServerApp] Shutting down 1 kernel
[I 2026-02-17 16:21:03.278 ServerApp] Kernel shutdown: 4172d409-f78f-45c3-8ead-f51
ac05fbde5
srliupp@vbox:~/project/notebooks/01_exponential_growth$ cd
srliupp@vbox:~$ cd /home/srliupp/work/study/2026-1/2026-1==study--mathmod/2026_1__
study_matmod/labs/lab01/report
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2026_1__study_matmod/labs/
lab01/report$ ls _quarto.yml
_quarto.yml
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2026_1__study_matmod/labs/
lab01/report$ gedit _quarto.yml
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2026_1__study_matmod/labs/
lab01/report$ gedit _quarto.yml
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2026_1__study_matmod/labs/
lab01/report$ cd
srliupp@vbox:~$ cd /home/srliupp/work/study/2026-1/2026-1==study--mathmod/2026_1__
study_matmod/labs/lab01/report/_resources/tex
srliupp@vbox:~/work/study/2026-1/2026-1==study--mathmod/2026_1__study_matmod/labs/
lab01/report/_resources/tex$
```

Рисунок 29: juliamono2

В каталоге отчёта в файл `_quarto.yml` включила поддержку кода `julia`. Добавила после описания проекта (рис. 30) (рис. 31) (рис. 32).

```
1 project:
2   title: "report"
3   output-dir: "_output"
4   standalone: true
5   self-contained: true
6
7 ## Julia support
8 engine: julia
9 julia:
10   exeflags: ["--project=../project"]
11
12 ## Generic options
13 lang: ru-RU
14 number-sections: true
15 toc: true
16 toc-title: "Содержание"
17 toc-depth: 2
18 ## Crossref customization
19 crossref:
20   lof-title: "Список иллюстраций"
21   lot-title: "Список таблиц"
22   lol-title: "Листинги"
23 ## Bibliography
24 bibliography:
25   - bib/cite.bib
26 csl: _resources/csl/gost-r-7-0-5-2008-numeric.csl
27 ## Formats
28 format:
```

Рисунок 30: quarto2



```
project:
  title: "report"
  output-dir: "_output"

standalone: true
self-contained: true

## Julia support
# engine: julia
# julia:
#   execlags: ["--project=../project"]

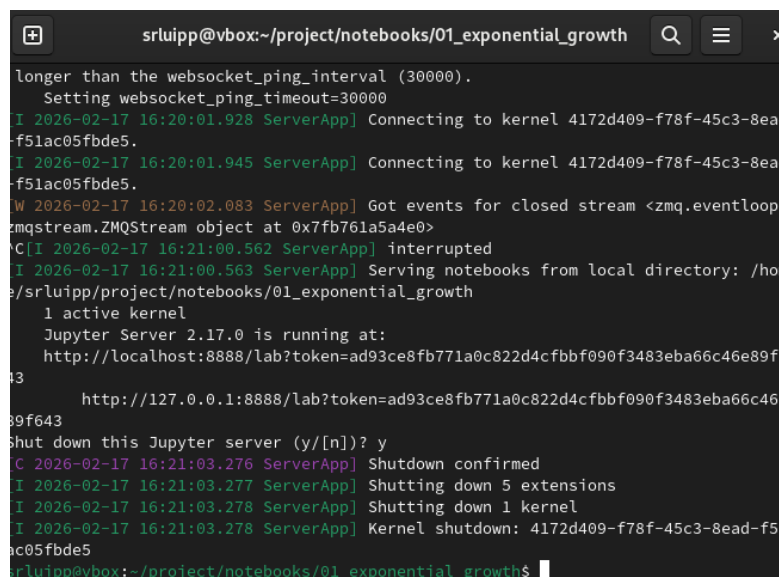
## Generic options
lang: ru-RU
number-sections: true
toc: true
toc-title: "Содержание"
toc-depth: 2
## Crossref customization
crossref:
  lof-title: "Список иллюстраций"
```

Рисунок 31: quarto3



Рисунок 32: quarto4

Устанавливаю jupyter (рис. 33).



```
srliupp@vbox:~/project/notebooks/01_exponential_growth
longer than the websocket_ping_interval (30000).
Setting websocket_ping_timeout=30000
[I 2026-02-17 16:20:01.928 ServerApp] Connecting to kernel 4172d409-f78f-45c3-8ead-f51ac05fbde5.
[I 2026-02-17 16:20:01.945 ServerApp] Connecting to kernel 4172d409-f78f-45c3-8ead-f51ac05fbde5.
[W 2026-02-17 16:20:02.083 ServerApp] Got events for closed stream <zmq.eventloop.ZMQStream object at 0x7fb761a5a4e0>
[C I 2026-02-17 16:21:00.562 ServerApp] interrupted
[I 2026-02-17 16:21:00.563 ServerApp] Serving notebooks from local directory: /home/srliupp/project/notebooks/01_exponential_growth
1 active kernel
Jupyter Server 2.17.0 is running at:
http://localhost:8888/lab?token=ad93ce8fb771a0c822d4cfbbf090f3483eba66c46e89f643
http://127.0.0.1:8888/lab?token=ad93ce8fb771a0c822d4cfbbf090f3483eba66c46e89f643
shut down this Jupyter server (y/[n])? y
[C 2026-02-17 16:21:03.276 ServerApp] Shutdown confirmed
[I 2026-02-17 16:21:03.277 ServerApp] Shutting down 5 extensions
[I 2026-02-17 16:21:03.278 ServerApp] Shutting down 1 kernel
[I 2026-02-17 16:21:03.278 ServerApp] Kernel shutdown: 4172d409-f78f-45c3-8ead-f51ac05fbde5
srliupp@vbox:~/project/notebooks/01_exponential_growth$
```

Рисунок 33: jupyter

Через терминал вывожу jupyter в mozilla firefox (рис. 34).

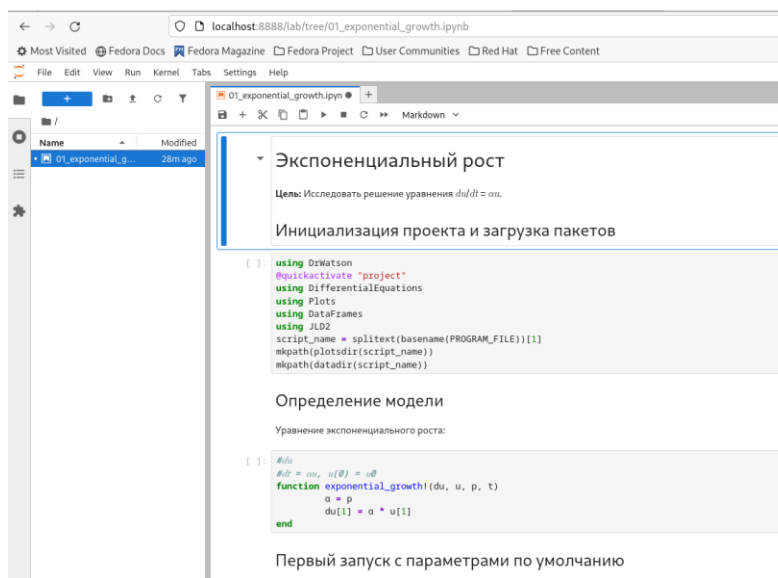


Рисунок 34: jupyter2

Генерация файла с экспоненциальным ростом через julia (рис. 35).

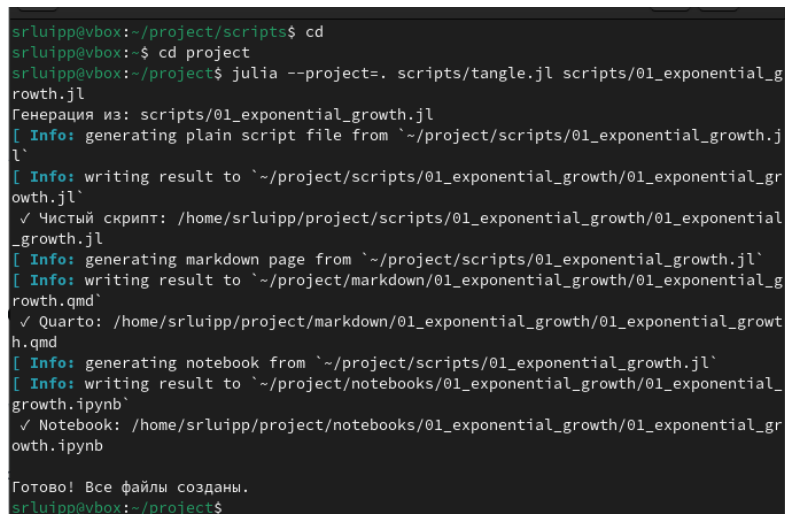


Рисунок 35: exponential

Презентация экспоненциального роста в папке plots (рис. 36).

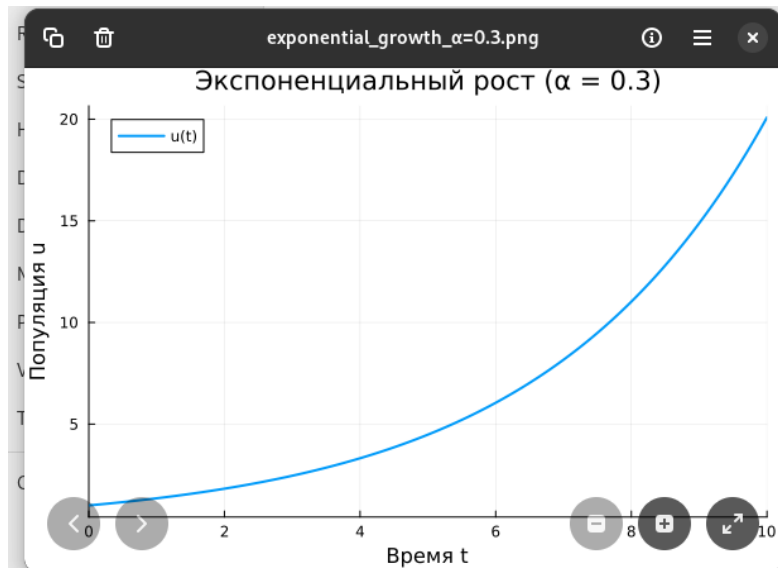


Рисунок 36: juliaplots

Вывод файла exponential (рис. 37) (рис. 38) (рис. 39) (рис. 40).

```
srluipp@vbox:~$ cd project
srluipp@vbox:~/project$ julia --project=. scripts/01_exponential_growth.jl
ERROR: LoadError: UndefVarError: `du` not defined
in expression starting at /home/srluipp/project/scripts/01_exponential_growth.jl:1:1
srluipp@vbox:~/project$ cd project/scripts
bash: cd: project/scripts: No such file or directory
srluipp@vbox:~/project$ cd scripts
srluipp@vbox:~/project/scripts$ gedit 01_exponential_growth.jl
srluipp@vbox:~/project/scripts$ cd
srluipp@vbox:~$ cd project
srluipp@vbox:~/project$ julia --project=. scripts/01_exponential_growth.jl
Первые 5 строк результатов:
5x2 DataFrame
 Row      t      u
   Float64 Float64
1      0.0  1.0
2      0.1  1.03045
3      0.2  1.06184
4      0.3  1.09417
5      0.4  1.1275
Аналитическое время удвоения: 2.31
srluipp@vbox:~/project$
```

Рисунок 37: result


```

rluipp@vbox:~/project$ julia --project=. scripts/01_exponential_growth.jl
Первые 5 строк результатов:
5x2 DataFrame
  Row  t      u
     Float64  Float64
1     0.0  1.0
2     0.1  1.03045
3     0.2  1.06184
4     0.3  1.09417
5     0.4  1.1275

Аналитическое время удвоения: 2.31
rluipp@vbox:~/project$ start plots\exponential_growth_α=0.3.png
bash: start: command not found...
C
rluipp@vbox:~/project130$ cd plots
rluipp@vbox:~/project/plots$ ls
exponential_growth_α=0.3.png'
rluipp@vbox:~/project/plots$ cd
rluipp@vbox:~$ cd project/script
bash: cd: project/script: No such file or directory
rluipp@vbox:~1$ cd project/scripts
rluipp@vbox:~/project/scripts$

```

Рисунок 38: results

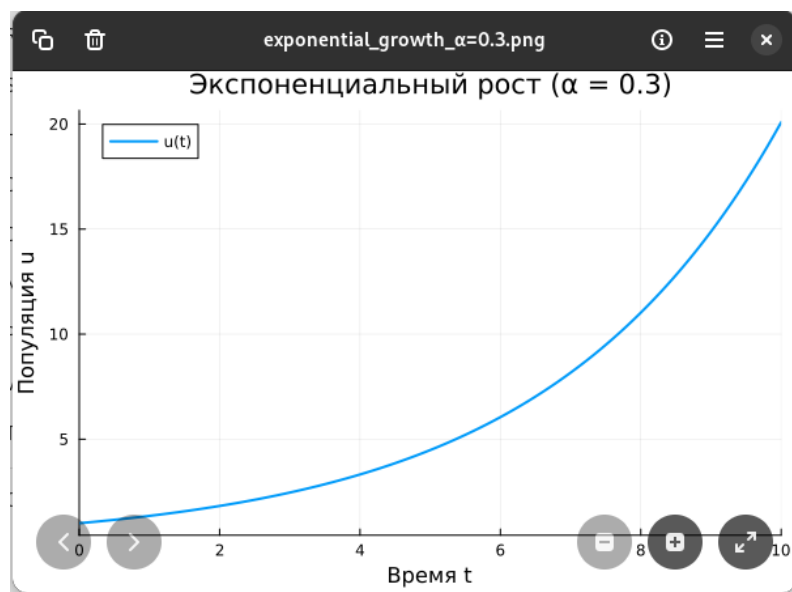


Рисунок 39: results1

```
✓ Literate
✓ IJulia
✓ BenchmarkTools
✓ Quarto

Структура проекта:
Корень: /home/srluipp/project
Данные: /home/srluipp/project/data
Скрипты: /home/srluipp/project/src
Графики: /home/srluipp/project/plots
srluipp@vbox:~/project$ julia --project scripts/01_exponential_growth.jl
Первые 5 строк результатов:
5×2 DataFrame
  Row    t      u
  Float64 Float64
1      0.0  1.0
2      0.1  1.03045
3      0.2  1.06184
4      0.3  1.09417
5      0.4  1.1275

Аналитическое время удвоения: 2.31
srluipp@vbox:~/project$
```

Рисунок 40: result2

Создаю новый файл для описания экспоненциального роста (рис. 41) (рис. 42).

```
srluipp@vbox:~/scripts$ cd
srluipp@vbox:~$ julia --project=. scripts/test_setup.jl
✓ Проект активирован: /home/srluipp

Проверка пакетов:
✓ DrWatson
✓ DifferentialEquations
✓ Plots
✓ DataFrames
✓ CSV
✓ JLD2
✓ Literate
✓ IJulia
✓ BenchmarkTools
✓ Quarto

Структура проекта:
Корень: /home/srluipp
Данные: /home/srluipp/data
Скрипты: /home/srluipp/src
Графики: /home/srluipp/plots
srluipp@vbox:~$
```

Рисунок 41: exp

```
Open ▾ 
1 using DrWatson
2 @quickactivate "project"
3 using DifferentialEquations
4 using Plots
5 using DataFrames
6 function exponential_growth!(du, u, p, t)
7     α = p
8     du[1] = α * u[1]
9 end
10 u0 = [1.0] # начальная популяция
11 α = 0.3 # скорость роста
12 tspan = (0.0, 10.0) # временной интервал
13 prob = ODEProblem(exponential_growth!, u0, tspan, α)
14 sol = solve(prob, Tsit5(), saveat=0.1)
15 plot(sol, label="u(t)", xlabel="Время t", ylabel="Популяция u",
16 title="Экспоненциальный рост (α = $α)", lw=2, legend=:topleft)
17 savefig(plotsdir("exponential_growth_α=$α.png"))
18 df = DataFrame(t=sol.t, u=first.(sol.u))
19 println("Первые 5 строк результатов:")
20 println(first(df, 5))
21
```

Рисунок 42: exp2

Устанавливаю пакеты DrWatson и другие (рис. 43) (рис. 44) (рис. 45)(рис. 46) .

```
[4ec0833e] + Unicode
[e66e0078] + CompilerSupportLibraries_jll v1.1.0+0
[deac9b47] + LibCURL_jll v8.4.0+0
[e37daf67] + LibGit2_jll v1.6.4+0
[29816b5a] + LibSSH2_jll v1.11.0+1
[c8ffd9c3] + MbedTLS_jll v2.28.2+1
[14a3606d] + MozillaCACerts_jll v2023.1.10
[4536629a] + OpenBLAS_jll v0.3.23+4
[05823500] + OpenLibm_jll v0.8.1+2
[efcefd77] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[83775a58] + Zlib_jll v1.2.13+1
[8e850b90] + libblastrampoline_jll v5.8.0+1
[8e850ede] + nghttp2_jll v1.52.0+1
[3f19e933] + p7zip_jll v17.4.0+2

Info Packages marked with ^ and ⚠ have new versions available. Those with ^
may be upgradable, but those with ⚠ are restricted by compatibility constraints from
upgrading. To see why use `status --outdated -m`
Precompiling project...
 1 dependency successfully precompiled in 5 seconds. 466 already precompiled.

✔ Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots
srlipp@vbox:~$
```

Рисунок 43: drwatson

```

srluipp@vbox:~$ julia add_packages.jl
Activating new project at `~`
Установка базовых пакетов...
Resolving package versions...
Updating `~/Project.toml`
[6e4b80f9] + BenchmarkTools v1.6.3
[336ed68f] + CSV v0.10.15
[a93c6f00] + DataFrames v1.8.1
[0c46a032] + DifferentialEquations v7.17.0
[634d3b9d] + DrWatson v2.19.1
[7073ff75] + IJulia v1.34.3
[033835bb] + JLD2 v0.6.3
[98b081ad] + Literate v2.21.0
[91a5bcd] + Plots v1.41.5
[d7167be5] + Quarto v1.0.0
Updating `~/Manifest.toml`
[47edcb42] + ADTypes v1.21.0
[7d9f7c33] + Accessors v0.1.43
[79e6a3ab] + Adapt v4.4.0
[66dad0bd] + AliasTables v1.1.3
[a95523ee] + AlmostBlockDiagonals v0.1.10
[ec485272] + ArnoldiMethod v0.4.0
[4fba245c] + ArrayInterface v7.22.0

```

Рисунок 44: *packages*

```

Open ▼ +
1 #!/usr/bin/env julia
2 ## add_packages.jl
3 using Pkg
4 Pkg.activate(".") # Активируем текущий проект
5 ## ОСНОВНЫЕ ПАКЕТЫ ДЛЯ РАБОТЫ
6 packages = [
7 "DrWatson", # Организация проекта
8 "DifferentialEquations", # Решение ОДУ
9 "Plots", # Визуализация
10 "DataFrames", # Таблицы данных
11 "CSV", # Работа с CSV
12 "JLD2", # Сохранение данных
13 "Literate", # Literate programming
14 "IJulia", # Jupyter notebook
15 "BenchmarkTools", # Бенчмаркинг
16 "Quarto" # Создание отчетов
17 ]

```

Рисунок 45: *packet*

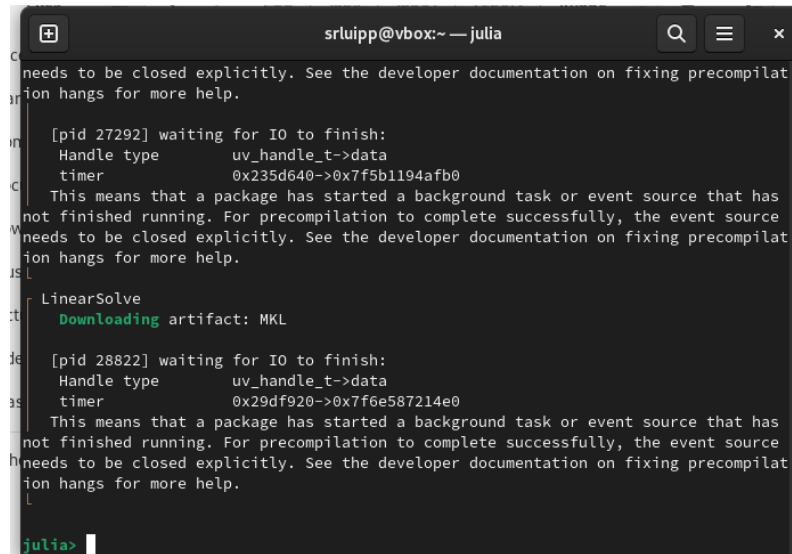
```

julia> println("\n✅ Все пакеты установлены!")
✅ Все пакеты установлены!
julia> println("Для проверки: using DrWatson, DifferentialEquations, Plots")
Для проверки: using DrWatson, DifferentialEquations, Plots
julia>

```

Рисунок 46: *doneinstall*

Работа с Julia (рис. 47).



```
needs to be closed explicitly. See the developer documentation on fixing precompil
ion hangs for more help.

[pid 27292] waiting for IO to finish:
Handle type      uv_handle_t->data
timer           0x235d640->0x7f5b1194afb0
This means that a package has started a background task or event source that has
not finished running. For precompilation to complete successfully, the event source
needs to be closed explicitly. See the developer documentation on fixing precompil
ion hangs for more help.

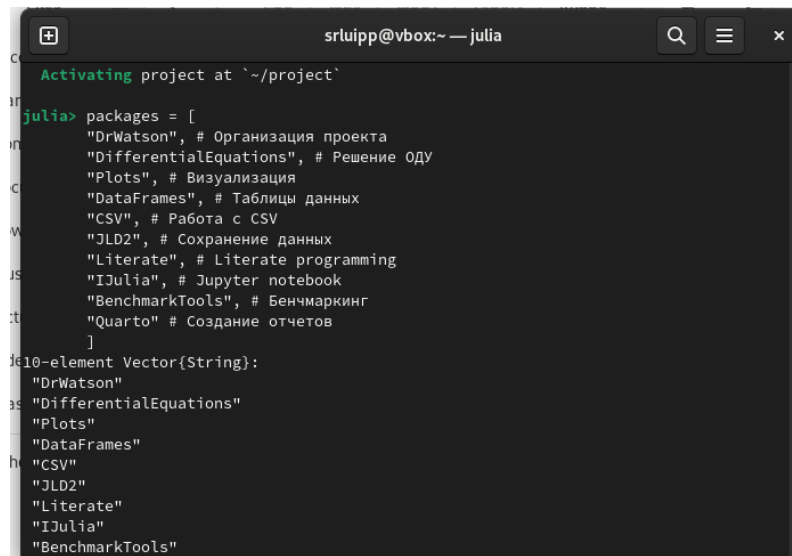
LinearSolve
  Downloading artifact: MKL

[pid 28822] waiting for IO to finish:
Handle type      uv_handle_t->data
timer           0x29df920->0x7f6e587214e0
This means that a package has started a background task or event source that has
not finished running. For precompilation to complete successfully, the event source
needs to be closed explicitly. See the developer documentation on fixing precompil
ion hangs for more help.

julia>
```

Рисунок 47: julia1

Установка пакетов для Julia (рис. 48) (рис. 49) (рис. 50) (рис. 51).

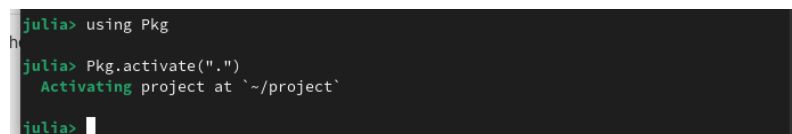


```
Activating project at `~/project`

julia> packages = [
  "DrWatson", # Организация проекта
  "DifferentialEquations", # Решение ОДУ
  "Plots", # Визуализация
  "DataFrames", # Таблицы данных
  "CSV", # Работа с CSV
  "JLD2", # Сохранение данных
  "Literate", # Literate programming
  "IJulia", # Jupyter notebook
  "BenchmarkTools", # Бенчмаркинг
  "Quarto" # Создание отчетов
]

10-element Vector{String}:
 "DrWatson"
 "DifferentialEquations"
 "Plots"
 "DataFrames"
 "CSV"
 "JLD2"
 "Literate"
 "IJulia"
 "BenchmarkTools"
```

Рисунок 48: juliapackages



```
julia> using Pkg
julia> Pkg.activate(".")
Activating project at `~/project`

julia>
```

Рисунок 49: juliapackwt

```

[83775a58] + Zlib_jll v1.2.13+1
[8e850ede] + nghttp2_jll v1.52.0+1
[3f19e933] + p7zip_jll v17.4.0+2
Info Packages marked with ⚠ have new versions available but compatibility co
nstraints restrict them from upgrading. To see why use `status --outdated -m`
Precompiling project...
18 dependencies successfully precompiled in 56 seconds. 1 already precompiled.

julia> initialize_project("project"; authors="Люпп София", git=false)
Activating new project at `~/project`
Resolving package versions...
Updating `~/project/Project.toml`
[634d3b9d] + DrWatson v2.19.1
Updating `~/project/Manifest.toml`
[0b6fb165] + ChunkCodecCore v1.0.1
[4c0bbe4] + ChunkCodecLibZlib v1.0.0
[55437552] + ChunkCodecLibZstd v1.0.0
[634d3b9d] + DrWatson v2.19.1
[5789e2e9] + FileIO v1.18.0
[076d061b] + HashArrayMappedTries v0.2.0
[033835bb] + JLD2 v0.6.3
[692b3bcd] + JLLWrappers v1.7.1
[1914dd2f] + MacroTools v0.5.16
[bac558e1] + OrderedCollections v1.8.1

```

Рисунок 50: juliaரச

```

julia> Pkg.add("Dr.Watson")
Installing known registries into `~/.julia`
ERROR: `Dr.Watson` is not a valid package name
Stacktrace:
 [1] pkgerror(msg::String)
   @ Pkg.Types ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/Types.jl:70
 [2] check_package_name
   @ ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:133 [inlined]
 [3] add(ctx::Pkg.Types.Context, pkgs::Vector{...}; preserve::Pkg.Types.PreserveLevel,
 platform::Base.BinaryPlatforms.Platform, kwargs::@Kwargs{...})
   @ Pkg.API ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:235
 [4] add(pkgs::Vector{Pkg.Types.PackageSpec}; io::Base.TTY, kwargs::@Kwargs{...})
   @ Pkg.API ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:159
 [5] add(pkgs::Vector{Pkg.Types.PackageSpec})
   @ Pkg.API ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:148
 [6] add
   @ ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:147 [inlined]
 [7] add(pkg::String)
   @ Pkg.API ~/julia-1.10.2/share/julia/stdlib/v1.10/Pkg/src/API.jl:146
 [8] top-level scope
   @ REPL[3]:1
Some type information was truncated. Use `show(err)` to see complete types.

julia>

```

Рисунок 51: error

Знакомство с julia через терминал (рис. 52).

```
julia> | Documentation: https://docs.julialang.org
      | Type "?" for help, "]"? for Pkg help.
      | Version 1.10.2 (2024-03-01)
      | Official https://julialang.org/ release
      |
julia>
```

Рисунок 52: *acquaintancejulia*

Загрузка Julia вручную (рис. 53).

```

$ curl -u root:root -O https://julialang-s3.julialang.org/bin/linux/x64/1.10/julia-1.10.2-linux-x86_64.tar.gz
--2026-02-17 00:21:29-- https://julialang-s3.julialang.org/bin/linux/x64/1.10/julia-1.10.2-linux-x86_64.tar.gz
Resolving julialang-s3.julialang.org (julialang-s3.julialang.org)... 146.75.54.49, 2a04:4e42:7d::561
Connecting to julialang-s3.julialang.org (julialang-s3.julialang.org)|146.75.54.49|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 169654402 (162M) [application/x-tar]
Saving to: 'julia-1.10.2-linux-x86_64.tar.gz'

julia-1.10.2-linux-x 100%[=====] 161.79M  9.67MB/s   in 22s

2026-02-17 00:21:57 (7.32 MB/s) - 'julia-1.10.2-linux-x86_64.tar.gz' saved [169654402/169654402]

```

Рисунок 53: *installjulia*

5. ВЫВОДЫ

В ходе лабораторной работы я приобрела практические навыки работы с системой управления версиями Git, языком программирования Julia.

Список литературы